

University of South Asia

Object Oriented Programming Lab

Project Proposal

TOPIC:

Fitness Tracking System

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1. Project Title Page

Title:
Fitness Tracking Management System

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2. Introduction to the Project

The Fitness Tracking Management System is a software application designed to help users record their daily exercise activities and track calories burned. It is implemented in C++ using core Object-Oriented Programming (OOP) principles such as Inheritance, Polymorphism, and Encapsulation.

The application allows users to log different types of workouts (Cardio and Strength) and automatically calculates energy expenditure based on specific formulas. This project demonstrates how OOP can be used to build a logical and efficient health management tool

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3. Objectives of the Project

- To create a functional Fitness Tracker using **C++** and **OOP concepts**.
- To implement **Inheritance** by creating derived classes (Cardio, Strength) from a base Workout class.
- To utilize **Polymorphism** for dynamic calorie calculation and display using virtual functions.
- To learn how to manage dynamic memory using **Vectors** and Pointers.
- To provide a clear and user-friendly **menu-driven interface**.

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4. Tools and Environment Used

- **Visual Studio Code (VS Code):** Integrated Development Environment.
- **MinGW (g++):** C++ Compiler.
- **Windows PowerShell/CMD:** For program execution.
- **Libraries Used:**
  - <iostream>: For Input/Output.
  - <vector>: For storing workout history.
  - <string>: For naming exercises.
  - <iomanip>: For formatting the output table.

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5. Project Description and Features

Main Functionalities:
- **Convert Input to Data:** Takes user input for duration and intensity/heart rate.
- **Cardio Calculation:** Calculates calories based on time and heart rate.
- **Strength Calculation:** Calculates calories based on time and intensity level.
- **Session History:** Displays a temporary list of all workouts added during the runtime.

Key Features:
- **Menu-driven interface:** Easy to navigate options.
- **Input validation:** Ensures the user enters correct numbers.
- **OOP Implementation:** proper use of parent and child classes.
- **Formatted Output:** Uses standard characters to display a clean history table.

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6. Technical Architecture and Logic

Input Phase:
- User chooses an operation from the menu.
- Inputs **Duration** (in minutes) and specific attributes like **Heart Rate** or **Intensity**

Processing Phase:
- **Inheritance:** Cardio and Strength inherit common attributes from Workout.
- **Polymorphism:** The calculateCalories() function is overridden:
  - ***Cardio:*** Uses a formula based on heart rate.
  - ***Strength:*** Uses a multiplier based on intensity (High/Medium/Low).
- **Storage:** Objects are stored in a vector<Workout*> list.

Output Phase:
- Display the result using <iostream> and <iomanip>.
- Shows a summary table of the session.

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7. Flow of Execution

1. **Display Menu** (Add Cardio, Add Strength, View History, Exit).
2. **Get User Input**.
3. **If Cardio** → Create object, calculate calories, store in vector.
4. **If Strength** → Create object, check intensity, calculate calories, store in vector.

5. **If View History** → Iterate through vector and display formatted table.
6. **Ask to continue or exit**.

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8. Division of Work Among Members

The following table:

Member	Responsibilities
Muhammad Umair	Designed the Base Class Workout and Cardio derived class. Implemented the core formula logic.
Isbah Rizwan	Built the Strength class logic, implemented Intensity checks, and handled input validation.
Sahar Zaheer	Managed the FitnessTracker manager class, created the Main Menu UI, and formatted the output table.
All Members	Collaborated on debugging, logic flow, and final documentation.

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9. Challenges Faced During Development

- **Polymorphism Implementation:** Understanding how to store different child objects in a single list using pointers.
- **Garbage Values:** Initially faced issues with random numbers appearing in output (Solved by proper Constructors).
- **Formatting Issues:** Aligning the history table columns perfectly using setw without special characters.
- **Input Handling:** Ensuring the program doesn't crash if a user enters text instead of numbers.

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10. Learning Outcomes

- Proficiency in **C++ Class structures** and **Inheritance**.
- Hands-on experience with **Polymorphism** and Virtual Functions.
- Understanding of **Encapsulation** and data protection.
- Efficient debugging and logical problem solving.
- Improved teamwork and code coordination.

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11. Future Scope and Enhancements

- **File Handling:** Adding support to save data permanently to a text file.
- **BMI Calculator:** Integrating a Body Mass Index calculator.
- **Graphical Interface:** Creating a visual version using a GUI library.
- **User Accounts:** Adding login functionality for multiple users.

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12. Conclusion

The **Fitness Tracking Management System** project has helped the team explore the depth of **Object-Oriented Programming**. It efficiently demonstrates how real-world objects (like workouts) can be modeled in code using classes. The tool is educational, lightweight, and demonstrates the core power of C++ in building management systems.

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13. References and Tools

- VS Code Documentation
- C++ Reference (cplusplus.com)
- Course Notes and Lectures on OOP